

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 What are main problems in mechanical, electrical and pneumatic components in CNC machine?
- Q.24 Write a short note on automatic tool changer and also explain its working.
- Q.25 a) Write note on recirculating ball screw and nut assembly and its application.
b) Write note on servo motor and stepper motor.

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Roll No.

3rd Year / Advance Diploma in Tool and Die Making

Subject : CNC Technology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 CNC stands for_____.
- a) Computer Numerical Control
 - b) Computer Numerically Controlled
 - c) Computer Network Control
 - d) Computer Navigation Centre
- Q.2 CIM stands for_____.
- a) Computer Intelligent Machine
 - b) Computer Information Management
 - c) Computer Integrated Manufacturing
 - d) Computer Interface Management

- Q.3 What is the function of M06?
- a) End of tape b) End of program
 - c) End of block d) End of operation

- Q.4 A robot arm is also known as _____.
- a) Actuator b) Sensor
 - c) Drive d) All of the above

- Q.5 G-codes are also known as _____.
- a) Graphical Codes
 - b) Geometric Codes
 - c) Miscellaneous Codes
 - d) Geometry Codes

- Q.6 What is the code for coolant off?
- a) G09 b) G02
 - c) M01 d) M09

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define Tachometer.
- Q.8 Define Feedback system.
- Q.9 Give binary equivalent of 27.

- Q.10 Write full form of MCU.
- Q.11 Which code is use for linear interpolation movement.
- Q.12 Name any two robot components.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What are different types of robotic joints?
- Q.14 Explain cutter radius compensation and tool offset.
- Q.15 What are different types of slideways? Explain.
- Q.16 Explain the laws of robot.
- Q.17 Differentiate between open loop and close loop control.
- Q.18 Write advantage and disadvantages of CNC machine.
- Q.19 What are main advantage of LVDT.
- Q.20 What is FMS? What are its main components?
- Q.21 What are environmental conditions required for working of CNC machine?
- Q.22 Define Do Loops and Subroutine.